

TEAM X:
CAN SEGMENT ANYTHING REALLY SEGMENT ANYTHING?
A FAILURE ANALYSIS OF SAM ON DIFFICULT OBJECT SEGMENTATION CASES

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1. Task and Motivation

Task Statement:

In this project, we want to study how well the Segment Anything Model (SAM) can segment objects in different difficult cases. We are not mainly trying to train a new model. Our main aim is to test SAM properly and understand where it works good and where it fails.

Motivation:

SAM is a very popular segmentation model because it can create object masks using simple prompts like a point or a bounding box. But in real-world images, objects are not always clear. Some objects are small, some are very thin, some are overlapping with other objects, and sometimes object boundary is not clear. So before using SAM for annotation or real computer vision systems, we should understand its limitations properly.

Related Work:

SAM was introduced as a general segmentation model which can segment many different objects using prompts. Datasets like COCO, Pascal VOC, and LVIS already provide ground-truth segmentation masks, so they are useful for checking how good the predicted masks are. Many works show that SAM is strong, but in our project we focus more on finding its failure cases in a structured way.

Challenge:

The main challenges are:

- (1) SAM may work well for big and clear objects, but may fail for small objects.
- (2) SAM may miss thin parts like bicycle spokes, chair legs, handles, or wires.
- (3) In crowded scenes, SAM may merge multiple nearby objects into one mask.
- (4) If object and background look similar, SAM may include background also.
- (5) Different prompt types like box prompt and point prompt may give different results.

2. Goals

Challenges to Address:

In this project, we want to answer these questions:

- (1) On which type of objects does SAM perform well?
- (2) On which type of objects does SAM fail more?
- (3) Is box prompt better than point prompt for object segmentation?
- (4) What are the common failure types of SAM?

Planned Work:

- (1) Select a small but useful subset from COCO or Pascal VOC dataset.
- (2) Divide objects into groups like small objects, thin objects, crowded objects, and normal objects.
- (3) Run SAM using bounding box prompts and point prompts.
- (4) Compare SAM output masks with the ground-truth masks.
- (5) Calculate segmentation scores like IoU and mIoU.
- (6) Show visual examples where SAM works well and where it fails.
- (7) Discuss what we can learn from these results.

3. Methods

Models and Frameworks:

- (1) We will use pretrained Segment Anything Model as the main model.
- (2) We will use COCO or Pascal VOC annotations for ground-truth masks and bounding boxes.
- (3) We will use Python, PyTorch, OpenCV, NumPy, and pycocotools for running experiments and evaluation.
- (4) If time is available, we may also compare with MobileSAM or automatic mask generation.

Method Explanation:

First, we take an image and one object from the dataset. Then we give SAM a prompt, like the ground-truth bounding box of that object. SAM will generate a predicted mask. After that, we compare this predicted mask with the real ground-truth mask from the dataset. This helps us check how close SAM's output is to the correct segmentation.

We start with bounding box prompts because it is a fair way to test mask quality. In this case, SAM already knows the object location, so we are mainly checking how well it can create the correct mask. Then we can also test point prompts to see whether SAM becomes weaker when the prompt is less detailed.

Key Difference from Just Running SAM:

This project is not only about running SAM on some images. We are doing a proper analysis. We group objects based on difficulty and compare the results. We also study different failure types like over-segmentation, under-segmentation, merging objects, missing thin parts, and boundary confusion.

Computational Budget:

We mostly need only inference with pretrained SAM, so we do not need heavy training. One GPU from the university cluster should be enough.

4. Datasets

Dataset Selection:

- (1) **COCO val2017**: This will be our main dataset because it has many object categories, instance masks, bounding boxes, and real-world scenes.
- (2) **Pascal VOC Segmentation**: This can be used as a smaller and simpler dataset if needed.
- (3) **LVIS subset**: This can be used optionally if we want more rare and difficult object categories.

5. Evaluation

Evaluation Method:

We evaluate SAM by comparing its predicted mask with the ground-truth mask. For every object, we calculate how much the predicted mask overlaps with the real mask. We report both overall results and group-wise results. We also show visual examples because segmentation failures are easier to understand visually.

Metrics:

- (1) **IoU**: Measures overlap between predicted mask and ground-truth mask.
- (2) **mIoU**: Average IoU over all objects.
- (3) **Success Rate at IoU > 0.5**: Percentage of

objects where SAM gives a good enough mask.

(4) **Failure Rate at IoU < 0.3**: Percentage of objects where SAM clearly fails.

(5) **Group-wise mIoU**: Separate mIoU for small, thin, crowded, and normal objects.

Failure Analysis:

We will manually check some failure cases and divide them into these types:

(1) **Over-segmentation**: SAM includes background or nearby object also.

(2) **Under-segmentation**: SAM segments only part of the object.

(3) **Object merging**: SAM combines multiple objects into one mask.

(4) **Thin-part failure**: SAM misses thin parts like legs, handles, wires, or spokes.

(5) **Boundary confusion**: SAM fails when object and background look similar.

Expected Result:

We expect SAM to work well on large and clear objects. But for small objects, thin objects, crowded scenes, and unclear boundaries, the performance may drop. The final result will show where SAM can be trusted and where human correction or better prompts are still needed.

6. References

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